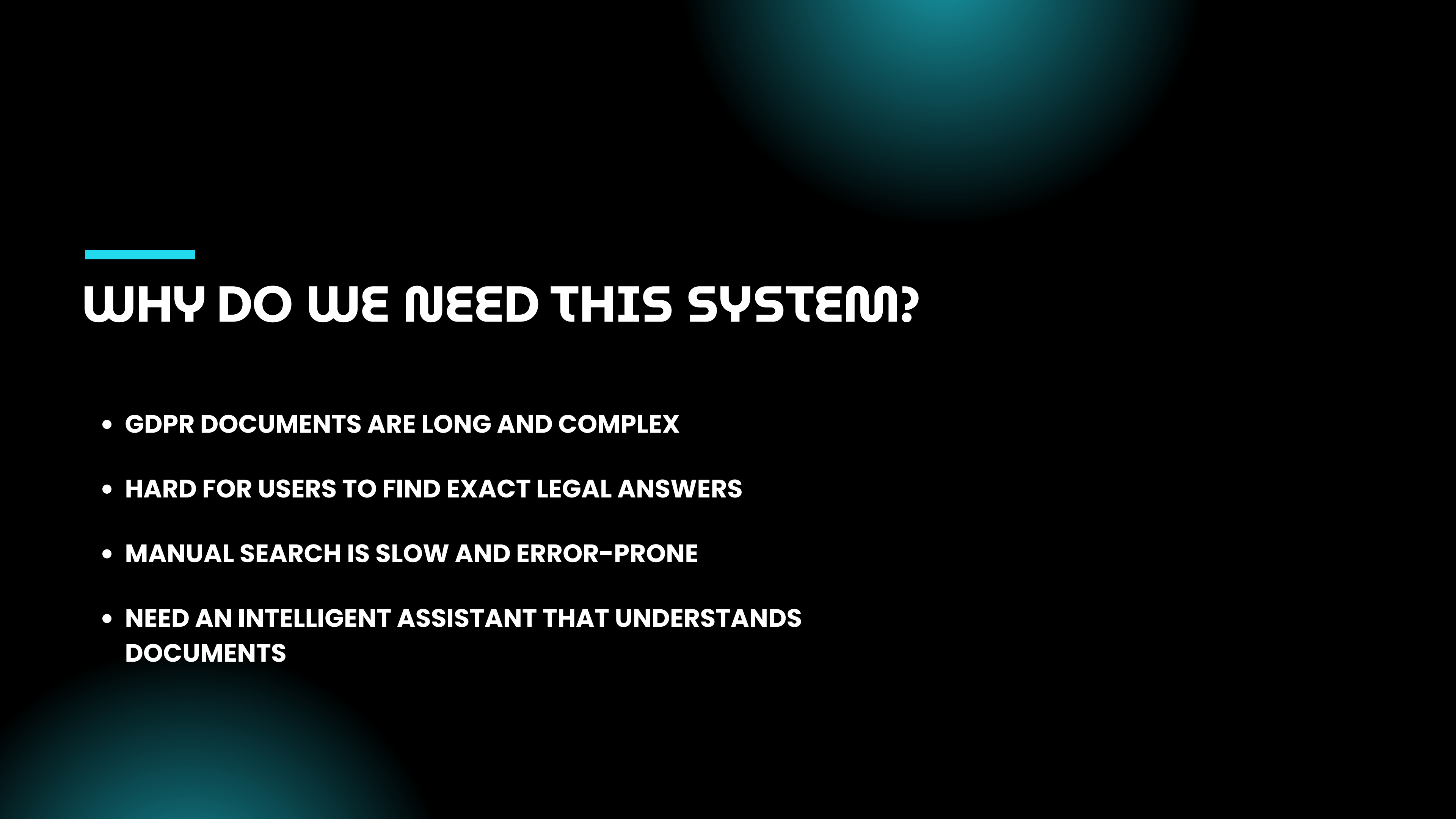


POLICYPULSE – QDPR RAQ SYSTEM

RETRIEVAL-AUGMENTED GENERATION
FOR LEGAL QUESTION ANSWERING



WHY DO WE NEED THIS SYSTEM?

- **GDPR DOCUMENTS ARE LONG AND COMPLEX**
- **HARD FOR USERS TO FIND EXACT LEGAL ANSWERS**
- **MANUAL SEARCH IS SLOW AND ERROR-PRONE**
- **NEED AN INTELLIGENT ASSISTANT THAT UNDERSTANDS DOCUMENTS**

SYSTEM ARCHITECTURE

**PDF → CHUNKING → EMBEDDINGS → VECTOR DB → RETRIEVAL → LLM → ANSWER + FACT
CHECK**

**PYPDFLOADER LOAD DOCUMENT
TOKENTEXTSPLITTER CHUNKING
HUGGINGFACE EMBEDDINGS VECTOR REPRESENTATION
CHROMADB STORAGE
GROQ LLM ANSWER GENERATION**

STEP 1: PDF PROCESSING

- PDF is loaded using PyPDFLoader
- Text is split into chunks:
 - chunk_size = 500 tokens
 - overlap = 50 tokens
- Each chunk gets metadata:
 - chunk_id
 - page number

STEP 2: EMBEDDING & STORAGE

- Model: all-MiniLM-L6-v2
- Converts text → vector embeddings
- Stored in: ChromaDB
- Similarity metric: Cosine similarity
- Why Chroma?
 - Fast retrieval
 - Persistent storage

WHEN USER ASKS A QUESTION:

1 Query is embedded

2 Similar chunks retrieved
(top 5)

3 Cosine similarity computed

4 Filtered by threshold (≥ 0.4)

5 Only relevant chunks
passed to LLM

LLM ANSWER GENERATION

1

Prompt rules:

- Use ONLY retrieved context
- Do NOT hallucinate
- Must cite GDPR articles if present
- If no info → say "Not found in retrieved GDPR text"

Fact Checking Layer

Extra safety layer

Second LLM call:

Input: Answer + Context

Output:

VERIFIED ---> fully supported

WARNING ---> possible hallucination

CONCLUSION

PolicyPulse = RAG-based GDPR assistant
Combines retrieval + generation + verification
Helps users understand legal text quickly and reliably

POLICYPULSE – GDPR RAQ SYSTEM

rana yasser
hazem mahmoud
mahmoud khamis
ahmed amr
badr ahmed
rahaf ehab